Determinant constructions for the Dirichlet beta value at four

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Abstract

The determinant proof that Catalan's constant is irrational compares, for a sequence of determinants with entries in the rationals and that constant, a lower bound from denominators with an upper bound from an integral representation. We record attempts to carry the method to the Dirichlet beta value at four. Several of them fail within the families tested. For determinants of half-integer arrays we measure both bounds up to size forty: the gap between them is four to six times the gap of the same construction for Catalan's constant, and the excess comes from the larger denominators of weight four. A weighted arcsine measure gives determinants with entries in the rationals and the beta value at four, with proofs, but they exceed the Catalan determinants with the same rows. We also measure determinants with poles at the negative squares, the construction of Fauzan's proof that the zeta value at five is irrational; their analogue for the beta value stays positive. Together the attempts point to a trade-off: the constructions with small denominators tie the beta value to a polylogarithmic constant, and the constructions that separate it have much larger denominators.

1 Introduction

Let $\beta(s) = \sum_{k\geq 0}(-1)^k(2k+1)^{-s}$, so that $G = \beta(2)$. Rivoal and Zudilin [2] proved, with very-well-poised hypergeometric series, that infinitely many of the values $\beta(2), \beta(4), \beta(6), \ldots$ are linearly independent over $\mathbb{Q}$ and that at least one of $\beta(2), \ldots, \beta(14)$ is irrational; no single even value beyond $\beta(2)$ is known to be irrational. We record some attempts to carry the determinant method of [1] to $\beta(4)$. Some fail for structural reasons and others are measured; the outcomes point to one trade-off. Unless a proof is given, the statements below are numerical observations (identifications by the PSLQ algorithm at 70 to 130 digits; exact rational arithmetic and determinants at up to 6460 digits where stated). They concern only the families tested. The method and its extension to general parameters are described in [1] and [9]; we use the notation of [1] and refer to its numbered equations as, for example, [1, Eq. (6)].

The attempts are the following: logarithmic kernels with the measure of [1] (Section 2), a second measure with rational moments (Section 3), the very-well-poised linear forms of Rivoal and Zudilin [2] (Section 4), determinants of half-integer arrays (Section 5), a weighted arcsine measure whose determinants have entries in $\mathbb{Q} + \mathbb{Q}\beta(4)$ (Section 6, Proposition 16), Hankel determinants of very-well-poised forms (Section 8), a scan of hypergeometric recurrences (Section 10), and determinants with poles at the negative squares, as in Fauzan's proof that $\zeta(5)$ is irrational [4] (Section 11). The first three fail within the families tested. For the half-integer determinants the gap between the two bounds levels off at four to six times the gap of the same construction for G (Observation 8), and the excess comes from the denominators of weight four (Observation 9); Sections 7 and 9 show that the denominators follow the weight. The determinants with poles at the negative squares stay near $+2$ per n^2 for $\beta(4)$, because the character modulo 4 halves the density of the poles relative to the decay

of the weight. Section 12 describes the common cause of the outcomes. Appendix A lists our checks of the proofs of [4] and [6], and Appendix B the scripts and runs. This note has not been refereed.

2 Logarithmic kernels with the arcsine measure

Replace the kernel $1/(1-ts)$ of [1, Eq. (6)] by $(-\log s)^r/(r!\,(1-ts))$ and keep the measure $|t|\,dt/\sqrt{1-t^2}$, whose moments are $m_{2k} = 2\cdot 4^k/((2k+1)\binom{2k}{k})$. The diagonal recurrences of [1, Sec. 2.1] carry over, and the boundary values $S_r(d) = \sum_u m_{d+u}/(u+1)^{r+1}$ satisfy

$$dS_r(d) = (d-1)S_r(d-2) + S_{r-1}(d-2) - S_{r-1}(d),$$

so every entry is a rational combination of 1 and of the starting values $S_r(0)$, $S_r(1)$ and their analogues on the other boundary, for orders up to r. With $t = \sin\theta$,

$$S_r(0) = \frac{2}{r!}\int_0^{\pi/2} \frac{\theta\,(-\log\sin\theta)^r}{\sin\theta}\,d\theta.$$

Observation 1. $S_0(0) = 4G$, $S_1(0) = 8\,\mathrm{Im}\,\mathrm{Li}_3\big(\frac{1+i}{2}\big) - \frac{\pi^3}{16} - \frac{\pi}{4}\log^2 2$, and

$$S_2(0) = -8\beta(4) + 16\,\mathrm{Im}\,\mathrm{Li}_4\Big(\frac{1+i}{2}\Big) + \frac{\pi^3\log 2}{16} + \frac{\pi\log^3 2}{12};$$

the other starting values contain only π^2, $\zeta(3)$, $\pi^2\log 2$, π^4, $\mathrm{Li}_4(\frac{1}{2})$ and powers of $\log 2$. More generally, every sum $\sum_k a_k R(k)$ with $a_k = 4^k/\binom{2k}{k}$ and R rational reduces, by partial fractions and $a_{k+1} = a_k(2k+2)/(2k+1)$, to the sums $A_n = \sum_k a_k/(2k+1)^n$ and $C_n = \sum_{k\ge 1} a_k/(2k)^n$; at weight four $A_4 = \frac{1}{2}S_2(0)$, while C_2, C_3, C_4 contain no β-type constant.

In this family $\beta(4)$ appears only in the fixed combination A_4, together with $\mathrm{Im}\,\mathrm{Li}_4((1+i)/2)$, and Taylor contact, which removes whole starting values, cannot separate them.

3 A second measure with rational moments

The measure $\arcsin(t)\,dt/\sqrt{1-t^2}$ also has rational moments, $p_n = \int_0^{\pi/2}\theta\sin^n\theta\,d\theta$ for odd n, with $n^2 p_n = n(n-1)p_{n-2} + 1$ and $p_1 = 1$. (The weight $\arcsin^2 t$ does not: its moments lie in $\mathbb{Q} + \mathbb{Q}\pi$, with π-part equal to p_n, so the π-part of every entry is a whole moment array, which no condition on the rows removes.)

Observation 2. For the measure $\arcsin(t)\,dt/\sqrt{1-t^2}$ the starting values carrying β-type constants are, for the kernel $1/(1-ts)$,

$$\tfrac{5\pi^3}{16} + \tfrac{\pi}{4}\log^2 2 - 4G\log 2 - 8\,\mathrm{Im}\,\mathrm{Li}_3\big(\tfrac{1+i}{2}\big),$$

and for the kernel $(-\log s)/(1-ts)$,

$$-28\beta(4) + 32\,\mathrm{Im}\,\mathrm{Li}_4\big(\tfrac{1+i}{2}\big) + 8\log 2\;\mathrm{Im}\,\mathrm{Li}_3\big(\tfrac{1+i}{2}\big) + \tfrac{\pi^2}{6}G + \tfrac{5\pi^3\log 2}{16} - \tfrac{\pi\log^3 2}{12};$$

the others are of ζ-type.

In the coordinates $(\beta(4), \mathrm{Im}\,\mathrm{Li}_4, \log 2\,\mathrm{Im}\,\mathrm{Li}_3, \pi^2 G, \pi^3\log 2, \pi\log^3 2)$ the two weight-four starting values are $(-8, 16, 0, 0, \frac{1}{16}, \frac{1}{12})$ and $(-28, 32, 8, \frac{1}{6}, \frac{5}{16}, -\frac{1}{12})$. The vector $(1,0,0,0,0,0)$ is not in their span, and starting values of lower weight cannot supply the weight-four products. Combining the two measures does not isolate $\beta(4)$ either.

4 Very-well-poised linear forms

The series $S_n(z)$ of [2, Eqs. (5)–(7)] with $q = 6$ give, at $z = -1$, linear forms $c_0 + c_2\beta(2) + c_4\beta(4)$ with rational coefficients [2, Lemma 2]. We computed them by exact partial fractions; the exact forms agree with the directly summed series to all digits.

Observation 3. Write the forms with coprime integer coefficients. Their size grows: $\log|\cdot|/n$ equals 2.0, 2.9, 4.0, 4.7 at $n = 4, 8, 12, 16$ for the parameters $(3; 1, \dots, 1)$ of [2, Eq. (3)], and 2.8, 5.0, 5.2, 5.8 for $(5; 2, \dots, 2)$; the forms decay like $e^{-5.3n}$ and e^{-7n}, their denominators grow like e^{10n} and $e^{12.7n}$. Eliminating $\beta(2)$ between the two families gives forms that grow like e^{14n}.

A single form would need size tending to 0; these forms miss by a factor e^{5n} before $\beta(2)$ is removed.

5 Determinants of half-integer arrays

The constants can be separated with half-integer shifts and an alternating sign. For $m \geq 1$ let

$$Z'_m(i,j) = \sum_{u\geq 0} \frac{(-1)^u}{(i+u+\frac{1}{2})^m\,(j+u+\frac{1}{2})} = \int_0^1\!\!\int_0^1 \frac{t^{i-\frac{1}{2}}s^{j-\frac{1}{2}}}{1+ts}\,\frac{(-\log t)^{m-1}}{(m-1)!}\,ds\,dt,$$

and $T_m(j) = \sum_{u\geq 0}(-1)^u(j+u+\frac{1}{2})^{-m} = (-1)^j\big(2^m\beta(m) - \sum_{v<j}(-1)^v(v+\frac{1}{2})^{-m}\big)$, with $\beta(1) = \pi/4$ and $\beta(3) = \pi^3/32$.

Lemma 4. *Let $m \in \{1,2,3\}$. For $i \neq j$ put $d = i - j$. Then $Z'_m(j,j) = T_{m+1}(j)$ and*

$$Z'_m(i,j) = \frac{T_1(j) - T_1(i)}{d^m} - \sum_{l=2}^{m} \frac{T_l(i)}{d^{\,m-l+1}} \qquad (i \neq j).$$

For $m = 3$ this reads $Z'_3(i,j) = \frac{T_1(j)-T_1(i)}{d^3} - \frac{T_2(i)}{d^2} - \frac{T_3(i)}{d}$.

Proof. With $x = j + u + \frac{1}{2}$ and $y = i + u + \frac{1}{2} = x + d$, the partial-fraction identity

$$\frac{1}{y^m x} = \frac{1}{d^m x} - \frac{1}{d^m y} - \sum_{l=2}^{m} \frac{1}{d^{\,m-l+1}y^l}$$

holds for $m \geq 1$ (for $m = 1$ it is $\frac{1}{yx} = \frac{1}{d}(\frac{1}{x} - \frac{1}{y})$, and the step from m to $m+1$ follows by dividing by y and applying the case $m = 1$ to $\frac{1}{d^m yx}$). Multiply by $(-1)^u$ and sum over $u \geq 0$. The case $i = j$ is the definition of T_{m+1}. □

Thus $\beta(4)$ occurs only on the diagonal, as $\zeta(2)$ does in the array Z of [1, Eq. (17)], and the other constants have explicit kernels: π through $((-1)^j - (-1)^i)/d^3$, which vanishes when $i \equiv j \pmod 2$, and G, π^3 through $(-1)^i/d^2$, $(-1)^i/d$. Restricting rows and columns to even indices removes π, and the arrays Z'_1, Z'_2, whose diagonals carry G and π^3, absorb those two constants row by row, as D_r absorbs $\zeta(2)$ in [1]. Definition 5 makes this explicit.

Definition 5. Let $n \geq 1$ and $\alpha, h, b, q \geq 0$ be integers. For $0 \leq r, k < n$ define the integers $a^{(3)}_{r,i}$ and $c_{k,j}$ by

$$t^{2(\alpha+r)}(1-t^2)^h = \sum_i a^{(3)}_{r,i}\,t^i, \qquad s^{2(b+k)}(1-s^2)^q = \sum_j c_{k,j}\,s^j,$$

let I_r be the set of i with $a^{(3)}_{r,i} \neq 0$, and let J be the set of j with $c_{k,j} \neq 0$ for some k. All elements of I_r and J are even. For $j \in J$ put

$$a^{(2)}_{r,j} = \sum_{i \in I_r,\, i \neq j} \frac{a^{(3)}_{r,i}}{i-j}, \qquad a^{(1)}_{r,j} = \sum_{i \in J,\, i \neq j} \frac{a^{(2)}_{r,i}}{i-j} + \sum_{i \in I_r,\, i \neq j} \frac{a^{(3)}_{r,i}}{(i-j)^2}.$$

The *half-integer determinant of weight four* is $\Delta_n = \det(E_{r,k})_{0 \le r,k < n}$ with

$$E_{r,k} = \sum_{i \in I_r,\, j \in J} a^{(3)}_{r,i} c_{k,j} Z'_3(i,j) + \sum_{i,j \in J} \bigl(a^{(2)}_{r,i} c_{k,j} Z'_2(i,j) + a^{(1)}_{r,i} c_{k,j} Z'_1(i,j) \bigr),$$

and the *half-integer determinant of weight two* is $\Delta^{(2)}_n = \det(E^{(2)}_{r,k})_{0 \le r,k < n}$ with

$$E^{(2)}_{r,k} = \sum_{i \in I_r,\, j \in J} a^{(3)}_{r,i} c_{k,j} Z'_1(i,j).$$

Lemma 6. *There are rational numbers $\rho_{r,k}$, $\rho^{(2)}_{r,k}$ such that*

$$E_{r,k} = \rho_{r,k} + 16\,\beta(4) \sum_{j \in J} a^{(3)}_{r,j} c_{k,j}, \qquad E^{(2)}_{r,k} = \rho^{(2)}_{r,k} + 4\,G \sum_{j \in J} a^{(3)}_{r,j} c_{k,j}.$$

Proof. For even i, $T_l(i) = 2^l \beta(l) - \sum_{v<i} (-1)^v (v + \frac12)^{-l}$, so $T_l(i)$ is $2^l\beta(l)$ plus a rational number, and $T_1(j) - T_1(i)$ is rational for even i, j. Insert Lemma 4 into $E_{r,k}$. The constant $\beta(4)$ comes only from the diagonal of Z'_3, with coefficient $16 \sum_j a^{(3)}_{r,j} c_{k,j}$. The coefficient of $\beta(3)$ is $8 \sum_{j \in J} c_{k,j} \bigl(a^{(2)}_{r,j} - \sum_{i \neq j} a^{(3)}_{r,i}/(i-j) \bigr)$, from the diagonal of Z'_2 and the term $l = 3$ of Z'_3; it vanishes by the definition of $a^{(2)}$. The coefficient of $\beta(2) = G$ is $4 \sum_{j \in J} c_{k,j} \bigl(a^{(1)}_{r,j} - \sum_{i \neq j} a^{(2)}_{r,i}/(i-j) - \sum_{i \neq j} a^{(3)}_{r,i}/(i-j)^2 \bigr)$, from the diagonal of Z'_1 and the terms $l = 2$ of Z'_2 and Z'_3; it vanishes by the definition of $a^{(1)}$. The same computation with Z'_1 alone gives the statement for $E^{(2)}_{r,k}$. □

Hence the entries of Δ_n lie in $\mathbb{Q} + \mathbb{Q}\beta(4)$, those of $\Delta^{(2)}_n$ lie in $\mathbb{Q} + \mathbb{Q}G$, and each row is a single function of t, so the integral representation of [1, Sec. 5] applies.

Observation 7. The common denominator of the rational parts of $Z'_3(i,j)$, $i, j < H$ even, grows like e^{cH} with $c = 6.07,\ 7.25,\ 7.44$ at $H = 20, 40, 80$, compared with $2 \log \operatorname{lcm}(1, \dots, H)/H \approx 2$ for the arrays of [1].

The reason is that $\beta(4)$ alone needs the odd denominators $(2v+1)^4$ up to $2H$, while the arcsine measure brings G with denominators that are powers of 2 only (the moments m_{2k}).

The balance of the two bounds can be measured directly. By Lemma 6, $\Delta_n = P_n(\beta(4))$ for a polynomial $P_n \in \mathbb{Q}[x]$ of degree at most n. If $\beta(4) = p/w$ with integers p, w, $w \ge 1$, then $w^n \operatorname{den}(P_n)\, \Delta_n$ is an integer, where $\operatorname{den}(P_n)$ is the least common denominator of the coefficients; the integrality step of [1] with this multiplier needs $|\Delta_n|\, w^n \operatorname{den}(P_n) < 1$ for some n with $\Delta_n \neq 0$. Put

$$\lambda_n = \frac{\log |\Delta_n|}{n^2}, \qquad \varpi_n = -\frac{\log \operatorname{den}\bigl(P_n(\frac13)\bigr)}{n^2}, \qquad \operatorname{gap}_n = \lambda_n - \varpi_n,$$

and define $\lambda^{(2)}_n$, $\varpi^{(2)}_n$, $\operatorname{gap}^{(2)}_n$ in the same way from $\Delta^{(2)}_n$, with G replaced by $\frac13$. Since $\operatorname{den}(P_n(\frac13)) \le 3^n \operatorname{den}(P_n)$, we have $\log \operatorname{den}(P_n)/n^2 \ge -\varpi_n - (\log 3)/n$, so the integrality step needs $\operatorname{gap}_n < (\log 3 - \log w)/n \le (\log 3)/n$; for $\operatorname{gap}_n \ge (\log 3)/n$ it gives no contradiction at that n. In the design notation of this section the parameters are $(\alpha, h, b, q) = (\lfloor r_\alpha n \rceil, \lfloor r_h n \rceil, \lfloor r_b n \rceil, \lfloor r_q n \rceil)$ for fixed ratios $(r_\alpha, r_h, r_b, r_q)$, where $\lfloor x \rceil$ is the nearest integer, ties to the even one.

Observation 8 (Measured; $6 \le n \le 40$)**.** For the five ratio vectors in Table 1 and for $n \in \{6, 8, 10, 12, 14, 16, 20, 24, 28, 32, 40\}$, the determinants Δ_n, $\Delta_n^{(2)}$, $P_n(\frac{1}{3})$ and its weight-two analogue are nonzero, and both gaps are positive; $\mathrm{gap}_n \ge 2.96 > (\log 3)/6$ in every case. Both gaps increase slowly with n and level off. At $n = 40$ the ratio $\mathrm{gap}_n/\mathrm{gap}_n^{(2)}$ lies between 3.7 and 6.0, and $\mathrm{gap}_n - \mathrm{gap}_n^{(2)} \ge 4.09$, the minimum being attained at $(0, 0, 0, 0)$; across the five designs $\mathrm{gap}_{40}^{(2)}$ varies by 1.9. A least-squares fit $c_0 + c_1/n + c_2/n^2$ over $n \ge 12$ gives limits c_0 between 5.4 and 10.8 for weight four and between 0.95 and 2.8 for weight two. Every design with $h > 0$ or $q > 0$ has larger gaps, at both weights, than $(0, 0, 0, 0)$.

Table 1: Measured balance of the half-integer determinants (Observation 8). Values per n^2.

	weight four, $\beta(4)$			weight two, G		
$(r_\alpha, r_h, r_b, r_q)$	gap_{20}	gap_{40}	λ_{40}	$\mathrm{gap}_{20}^{(2)}$	$\mathrm{gap}_{40}^{(2)}$	$\lambda_{40}^{(2)}$
$(0, 0, 0, 0)$	4.48	4.91	−2.86	0.66	0.82	−2.99
$(0, \frac{1}{8}, 0, \frac{1}{8})$	5.68	6.40	−3.07	1.16	1.44	−3.23
$(0, \frac{1}{4}, 0, \frac{1}{4})$	7.50	8.05	−3.24	1.96	2.17	−3.43
$(0, 0, 0, \frac{1}{4})$	6.77	7.29	−3.12	1.37	1.53	−3.21
$(\frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4})$	10.23	10.77	−3.37	2.47	2.73	−3.87

The weight-two determinants of this family do not prove that G is irrational either: their gap is positive and increases with n. The proof in [1] uses a different construction (the arrays D_r and the arcsine measure). Observation 8 therefore compares the two weights within one family. The sizes λ_n and $\lambda_n^{(2)}$ are close; the difference lies in the denominators.

Observation 9 (Measured; exact arithmetic)**.** Replace every constant $\beta(1), \beta(2), \beta(3), \beta(4)$ by $\frac{1}{3}$ in the entries, and let ϖ denote $-\log \mathrm{den}(\det)/n^2$ of the resulting rational determinant. For the matrix of weight two, for the matrix of weight four built from the rows $a^{(3)}$ alone (so that G and π^3 remain in the entries), and for the matrix E of Definition 5, the values are:

$(r_\alpha, r_h, r_b, r_q)$, n	weight two	weight four, $a^{(3)}$ only	weight four, E
$(0, 0, 0, 0)$, 10	−2.98	−6.32	−6.08
$(0, 0, 0, 0)$, 20	−3.53	−7.28	−7.19
$(0, 0, 0, 0)$, 28	−3.68	−7.52	−7.50
$(0, 0, 0, 0)$, 40	−3.81	−7.76	−7.77
$(0, \frac{1}{4}, 0, \frac{1}{4})$, 10	−4.19	−9.09	−8.70
$(0, \frac{1}{4}, 0, \frac{1}{4})$, 20	−5.25	−10.78	−10.59
$(0, \frac{1}{4}, 0, \frac{1}{4})$, 28	−5.46	−11.12	−10.99
$(0, \frac{1}{4}, 0, \frac{1}{4})$, 40	−5.60	−11.39	−11.29

The cancellation of G and π^3 changes the exponent by less than 0.4 in every case and by less than 0.13 for $n \ge 28$, in either direction. The exponent at weight four is between 2.00 and 2.09 times the exponent at weight two (the values for $n = 14$ are in the log).

This is consistent with the source of the denominators: the rational parts of the entries contain the partial sums $\sum_{v<j}(-1)^v(v + \frac{1}{2})^{-4}$, whose denominators carry fourth powers of the odd numbers below $2j$, where weight two has squares.

6 A weighted arcsine measure

Sections 2 and 3 found $\beta(4)$ only together with $\operatorname{Im}\operatorname{Li}_4((1+i)/2)$. A measure of weight two whose boundary values contain $\beta(4)$ alone exists.

Definition 10. Let $q(\theta)=\theta^2-\pi\theta$ and let ν be the signed measure on $[-1,1]$ given by

$$d\nu(t)=|t|\,q(|\arcsin t|)\,\frac{dt}{\sqrt{1-t^2}}.$$

For integers $i,j\ge 0$ put $M_\nu(i,j)=\int_{-1}^{1}\int_0^1 t^i s^j(1-ts)^{-1}\,ds\,d\nu(t)$, and define $M_\mu(i,j)$ and $L_\mu(i,j)$ in the same way with μ in place of ν, the latter with the kernel $(-\log s)(1-ts)^{-1}$. Put $O_2(k)=\sum_{l=0}^{k}(2l+1)^{-2}$, $X_3=S_1(0)=\sum_{k\ge0}m_{2k}(2k+1)^{-2}$, and $W_n=\int_0^{\pi/2}\sin^n\theta\,d\theta$.

By Observation 1, $X_3=8\operatorname{Im}\operatorname{Li}_3(\frac{1+i}{2})-\frac{\pi^3}{16}-\frac{\pi}{4}\log^2 2$; this value is not used in the proofs below.

Lemma 11. *The odd moments of ν vanish, and $\nu_{2k}=\int t^{2k}\,d\nu=-2\,m_{2k}\,O_2(k)$ for $k\ge0$. In particular all moments are rational.*

Proof. The measure is even, so odd moments vanish. With $t=\sin\theta$, $\nu_{2k}=2I_{2k+1}$ where $I_n=\int_0^{\pi/2}q(\theta)\sin^n\theta\,d\theta$, and $m_{2k}=2W_{2k+1}$. For odd $n\ge1$ integrate by parts twice, using $(\sin^n)''=n(n-1)\sin^{n-2}-n^2\sin^n$, $q(0)=0$, $q'(\pi/2)=0$, $\sin^n(0)=0$, $(\sin^n)'(\pi/2)=0$ and $q''=2$:

$$n^2I_n=n(n-1)I_{n-2}-2W_n\qquad(n\ge3),\qquad I_1=-2W_1=-2.$$

Since $n^2W_n=n(n-1)W_{n-2}$, the quotient $\sigma_k=I_{2k+1}/W_{2k+1}$ satisfies $\sigma_k=\sigma_{k-1}-2(2k+1)^{-2}$ and $\sigma_0=-2$, so $\sigma_k=-2O_2(k)$. □

The boundary sequences are $K_d^-=M_\nu(d,0)=\sum_{u\ge0}\nu_{d+u}/(u+1)$ and $K_j^+=M_\nu(0,j)=\sum_{u\ge0}\nu_u/(u+j+1)$; $K_0^-=K_0^+$. For μ and the plain kernel they are the sequences of [1, Sec. 2.1]: $\kappa_d^-=M_\mu(d,0)\in\mathbb{Q}+4c(d/2)G$ for even d (and $\in\mathbb{Q}$ for odd d), $\kappa_j^+=M_\mu(0,j)\in\mathbb{Q}$ for even $j\ge2$, and $\kappa_0^\pm=S_0(0)=4G$.

Lemma 12. *$K_0^-=-8\beta(4)$, $K_2^+=-2X_3$, and $L_\mu(0,2)=4-4G$.*

Proof. First value. With $t=\sin\theta$, $K_0^-=\int(-\log(1-t))t^{-1}d\nu=2\int_0^{\pi/2}q(\theta)\operatorname{artanh}(\sin\theta)\,d\theta$, since ν is even and $-\log(1-t)+\log(1+t)=2\operatorname{artanh}t$. Put $R(\theta)=\theta^3/3-\pi\theta^2/2+\pi^3/12$, so that $R'=q$ and $R(\pi/2)=0$; integration by parts with $\frac{d}{d\theta}\operatorname{artanh}(\sin\theta)=\sec\theta$ gives $\int_0^{\pi/2}q\operatorname{artanh}(\sin\theta)\,d\theta=-\int_0^{\pi/2}R(\theta)\sec\theta\,d\theta$. With $\varphi=\pi/2-\theta$, $R=\frac{\pi^2}{4}\varphi-\frac{1}{3}\varphi^3$, hence

$$\begin{aligned}\int_0^{\pi/2}q\operatorname{artanh}(\sin\theta)\,d\theta&=-\frac{\pi^2}{4}\int_0^{\pi/2}\frac{\varphi}{\sin\varphi}d\varphi+\frac{1}{3}\int_0^{\pi/2}\frac{\varphi^3}{\sin\varphi}d\varphi\\&=-\frac{\pi^2}{4}\,2G+\frac{1}{3}\Big(\frac{3\pi^2}{2}G-12\beta(4)\Big)=-4\beta(4).\end{aligned}$$

The two integrals are classical: integrate by parts against $\log\tan(\varphi/2)=-2\sum_{k\ge0}\cos((2k+1)\varphi)/(2k+1)$ and integrate the series term by term.

Second value. By Lemma 11, $K_2^+=-2\sum_k m_{2k}O_2(k)/(2k+3)=-2\sum_l(2l+1)^{-2}\sum_{k\ge l}m_{2k}/(2k+3)$. From $m_{2k+2}=m_{2k}(2k+2)/(2k+3)$ we get $m_{2k}/(2k+3)=m_{2k}-m_{2k+2}$, and $m_{2k}\to0$, so the inner sum is m_{2l} and $K_2^+=-2X_3$.

Third value. $L_\mu(0,2)=\sum_k m_{2k}(2k+3)^{-2}=\sum_k(m_{2k}-m_{2k+2})(2k+3)^{-1}=\sum_{k\ge0}m_{2k}/(2k+3)-\sum_{k\ge1}m_{2k}/(2k+1)=m_0-(4G-m_0)=4-4G$, by the telescoping identity again and $S_0(0)=\sum_k m_{2k}/(2k+1)=4G$. □

Lemma 13. *Write* $e_n = -2m_n/(n+1)$, $P_d^1 = \sum_{n<d} m_n/(n+1)$, $P_d^2 = \sum_{n<d} m_n/(n+1)^2$ *and* κ_j^+ *as above. Then for* $d \geq 2$

$$dK_d^- = (d-1)K_{d-2}^- - \sum_{v<d-2} \frac{\nu_v}{v+3} - \frac{2}{d}(\kappa_d^- - 4G + P_d^1) - 2P_d^2,$$

and for $j \geq 1$

$$K_{j+2}^+ = \frac{jK_j^+ + K_2^+ - jF_j}{j+1}, \qquad F_j = -2\Big(\frac{X_3}{j} - \frac{4G}{j^2} + \frac{\kappa_j^+}{j^2}\Big).$$

For the array L_μ, *with* $\lambda_d^- = L_\mu(d,0)$ *and* $\lambda_j^+ = L_\mu(0,j)$: $\lambda_0^- = X_3$,

$$\lambda_d^- = \Big(1 - \frac{1}{d}\Big)\lambda_{d-2}^- + \frac{1}{d^2}\Big(\kappa_{d-2}^- - \kappa_2^+ + \sum_{v<d-2} \frac{m_v}{v+3}\Big) \quad (d \geq 2),$$

$$\lambda_{j+2}^+ = \frac{j}{j+1}\Big(\lambda_j^+ - \frac{m_0}{(j+1)^2} + \frac{\kappa_2^+ - \kappa_{j+2}^+}{j^2}\Big) \quad (j \geq 2).$$

All arrays satisfy $A(i,j) = A(i-1,j-1) - a_{i-1}/j^r$, *with* $a = \nu$, $r = 1$ *for* M_ν; $a = m$, $r = 1$ *for* M_μ; $a = m$, $r = 2$ *for* L_μ.

Proof. The last statement follows from $\int_0^1 s^{j-1}(1-ts)^{-1}ds - t\int_0^1 s^j(1-ts)^{-1}ds = 1/j$ and its analogue with $-\log s$. By Lemma 11, $(n+1)\nu_n = n\nu_{n-2} + e_n$ for even $n \geq 0$ (with $\nu_{-2} = 0$), and $(n+1)m_n = nm_{n-2}$. For K_d^- substitute $\nu_{d+u} = ((d+u)\nu_{d+u-2} + e_{d+u})/(d+u+1)$ and use $\frac{d+u}{(d+u+1)(u+1)} = \frac{1}{u+1} - \frac{1}{d}(\frac{1}{u+1} - \frac{1}{d+u+1})$; the sum $\sum_u e_{d+u}/((d+u+1)(u+1))$ is split with $\frac{1}{b^2a} = \frac{1}{d^2a} - \frac{1}{d^2b} - \frac{1}{db^2}$ ($a = u+1$, $b = d+u+1$), which produces κ_d^-, $S_0(0) - P_d^1$ and $X_3 - P_d^2$; the term $K_2^+ = -2X_3$ (Lemma 12) cancels X_3. The other three recurrences follow in the same way from the moment recurrences and partial fractions in the summation variable; for λ^+ the term $n = 0$ of the sum is kept apart because the recurrence for m_n starts at $n = 2$. □

Proposition 14. *For even* $i, j \geq 0$:

1. *if* $i \geq j$, *then* $M_\nu(i,j) \in \mathbb{Q} - 8c(\frac{i-j}{2})\beta(4) + \mathbb{Q}G$ *and* $L_\mu(i,j) \in \mathbb{Q} + c(\frac{i-j}{2})X_3 + \mathbb{Q}G$, $M_\mu(i,j) \in \mathbb{Q} + 4c(\frac{i-j}{2})G$;
2. *if* $i < j$, *then* $M_\nu(i,j) \in \mathbb{Q} + \mathbb{Q}G + \mathbb{Q}X_3$, $L_\mu(i,j) \in \mathbb{Q} + \mathbb{Q}G$, $M_\mu(i,j) \in \mathbb{Q}$.

Proof. Unrolling $A(i,j) = A(i-1,j-1) - a_{i-1}/j^r$ expresses $A(i,j)$ as $A(i-j,0)$ minus a rational sum if $i \geq j$, and as $A(0,j-i)$ minus a rational sum if $i < j$. By Lemma 13 and induction on d from $K_0^- = -8\beta(4)$ (Lemma 12), the K_d^- for even d lie in $\mathbb{Q} + \mathbb{Q}\beta(4) + \mathbb{Q}G$, the coefficient of $\beta(4)$ being multiplied by $(d-1)/d$ at each step, which gives $-8c(d/2)$. The sequence K_j^+ for even $j \geq 2$ starts from $K_2^+ = -2X_3$ and the recurrence adds only X_3, G and rationals; $K_0^+ = -8\beta(4)$ enters only the diagonal $i = j$, which belongs to case 1. The statements for L_μ follow in the same way from $\lambda_0^- = X_3$, $\lambda_2^+ = 4 - 4G$, and those for M_μ from the facts on $\kappa^\pm$ recalled above. □

Hence, for even indices, the three constants that the odd terms of $K^\pm$ carry (π^2, G and $G^2 - \pi^4/64$, observed numerically) never enter, and two auxiliary arrays suffice to remove X_3 and G.

Definition 15. Let $n \geq 1$ and $h, q \geq 0$. For $0 \leq r, k < n$ let $a_{r,i}$ and $c_{k,j}$ be the coefficients of $t^{2r}(1-t^2)^h = \sum_i a_{r,i}t^i$ and $s^{2k}(1-s^2)^q = \sum_j c_{k,j}s^j$, and let J be the set of j with $c_{k,j} \neq 0$ for some

k. Define rational numbers $a^L_{r,i'}$ and $a^\mu_{r,i'}$, $i' \in J$, by the triangular systems (solved from the largest j downwards)

$$\sum_{i' \in J,\, i' \ge j} a^L_{r,i'}\, c\big(\tfrac{i'-j}{2}\big) = -[X_3] \sum_i a_{r,i} M_\nu(i,j),$$

$$\sum_{i' \in J,\, i' \ge j} 4a^\mu_{r,i'}\, c\big(\tfrac{i'-j}{2}\big) = -[G]\Big(\sum_i a_{r,i} M_\nu(i,j) + \sum_{i'} a^L_{r,i'} L_\mu(i',j)\Big)$$

for all $j \in J$, where $[C]x$ denotes the coefficient of C in x. The entries are

$$E_{r,k} = \sum_j c_{k,j}\Big(\sum_i a_{r,i} M_\nu(i,j) + \sum_{i' \in J} a^L_{r,i'} L_\mu(i',j) + \sum_{i' \in J} a^\mu_{r,i'} M_\mu(i',j)\Big),$$

and $\Delta^\nu_n = \det(E_{r,k})_{0 \le r,k < n}$.

Proposition 16. *Every entry $E_{r,k}$ lies in $\mathbb{Q} + \mathbb{Q}\beta(4)$.*

Proof. By Proposition 14, X_3 occurs in $L_\mu(i',j)$ only for $i' \ge j$, with coefficient $c((i'-j)/2)$, and not in M_μ; the first system therefore removes X_3 column by column. After that, G occurs in $M_\mu(i',j)$ only for $i' \ge j$ with coefficient $4c((i'-j)/2)$, and the second system removes G; M_μ adds no other constant. Both systems are triangular with diagonal entries $c(0) = 1$ and 4, hence uniquely solvable. □

Observation 17 (Measured; $4 \le n \le 32$)**.** With the quantities of Section 5 ($\beta(4)$, respectively G, replaced by $\frac{1}{3}$ in the denominator), the gap of Δ^ν_n for $h = q = 0$, the gap of the Catalan determinant $\det(M_\mu(2r, 2k))$ with the same rows and columns, and the gap of the determinant of [1] at the ratios $48 : 7 : 4 : 4 : 2$ are:

n	8	12	16	20	24	28	32
Δ^ν_n, $\beta(4)$	2.47	2.94	3.19	3.41	3.47	3.63	3.68
$\det M_\mu$, G, same rows	0.01	0.10	0.14	0.21	0.26	0.32	0.34
[1], G	–	−0.30	–	−0.22	–	−0.15	–

At $n = 48$ the determinant of [1] has gap -0.12. For $h = q = n/4$ the gap of Δ^ν_n is 3.89, 4.38, 4.71 at $n = 8, 12, 16$. In every case the entries contain no constant other than $\beta(4)$ (exact check), and the determinants are nonzero.

The gap of Δ^ν_n is about 1.1 below that of the half-integer determinants of Observation 8 at the same n (4.26, 4.48, 4.64 at $n = 16, 20, 24$) and levels off near 3.7. At $n = 32$ the size is $\lambda = -1.74$ and the denominator exponent $\varpi = -5.42$; for the Catalan determinant with the same rows they are -2.05 and -2.39. The difference lies in the denominators.

7 Denominators prime by prime

For a determinant Δ with entries in $\mathbb{Q} + \mathbb{Q}C$ let D be the denominator of Δ with C replaced by $\frac{1}{3}$, as in Section 5. For a prime p write $v_p(D)$, the *row bound* $\sum_r \max_k v_p(\operatorname{den} E_{r,k})$, and the largest v_p of a single entry.

Observation 18 (Measured; exact arithmetic; $n = 16$)**.** For Δ^ν_{16} (weight four), for the Catalan determinant with the same rows and columns, and for the determinant of [1] at the ratios $48 : 7 : 4 : 4 : 2$ (both weight two), the triples ($v_p(D)$, row bound, largest entry) are:

	$p=3$	$p=5$	$p=7$	$p=11$	$p=13$
Δ_{16}^{ν}, $\beta(4)$	$(142,174,12)$	$(86,118,8)$	$(62,64,4)$	$(63,64,4)$	$(60,64,4)$
same rows, G	$(70,78,6)$	$(37,54,4)$	$(30,32,2)$	$(31,32,2)$	$(28,32,2)$
[1], G	$(61,64,4)$	$(31,32,2)$	$(27,32,2)$	$(28,32,2)$	$(20,32,2)$

Split by the size of the prime, $\log D/n^2$ is

p/n in	$[0,\frac{1}{4})$	$[\frac{1}{4},\frac{1}{2})$	$[\frac{1}{2},\frac{3}{4})$	$[\frac{3}{4},1)$	$[1,\frac{5}{4})$	$\geq\frac{5}{4}$	total
Δ_{16}^{ν}, $\beta(4)$	0.98	1.01	0.59	0.60	1.07	0.50	4.75
same rows, G	0.41	0.46	0.29	0.28	0.43	0.19	2.06
[1], G	0.48	0.40	0.26	0.20	0.11	0	1.45

At every odd prime the largest power in one entry doubles from weight two to weight four, and the valuation of the determinant about doubles. For $7 \leq p \leq 13$ both determinants with simple rows stay within four of the row bound, so almost nothing cancels. The rows of [1] reduce the valuation mainly at the large primes: their indices stop near $1.35n$, so primes above $1.3n$ do not occur. At the small primes ($p < n/2$) the two weight-two determinants are equal (0.88 and 0.87). The cancellations of the determinant method reduce the number of rows that carry a prime; the power per row is set by the weight.

8 Hankel determinants of Rivoal–Zudilin moments

The forms of Section 4 are small because their rational function vanishes at many integers. A determinant built from them might gain on the scale n^2, as in [1].

Definition 19. For a parameter vector η and $N \geq 1$ let R_N be the rational function of Section 4, $h_0 = \eta_0 N + 1$ and $u = t + h_0/2$. For $j \geq 0$ put $\mu_j(N) = \sum_{t\geq 0}(-1)^t R_N(t)\, u^{2j}$ whenever the degree of $R_N u^{2j}$ is negative, and $H_m = \det(\mu_{r+k}(N))_{0\leq r,k<m}$.

The factor u^{2j} is invariant under $t \mapsto -h_0 - t$, and in every case computed the coefficients of $\beta(1)$, $\beta(3)$ and $\beta(5)$ in $\mu_j(N)$ vanish (exact partial fractions), so $\mu_j(N) \in \mathbb{Q} + \mathbb{Q}\beta(2) + \mathbb{Q}\beta(4)$. For the test below both constants are replaced by $\frac{1}{3}$, which asks less than a proof for $\beta(4)$ would.

Observation 20 (Measured; exact arithmetic)**.** Let $B = \log|H_m| + \log \mathrm{den}(H_m)$ with $\beta(2), \beta(4) \to \frac{1}{3}$ in the denominator; a contradiction needs $B < 0$. For $\eta = (3; 1, 1, 1, 1, 1; 1)$ and $N = 12$,

m	1	2	3	4	6	8	10
B	$+47.5$	$+66.6$	$+79.4$	$+85.0$	$+78.3$	$+70.6$	$+89.5$

For $\eta = (5; 2, 2, 2, 2, 2; 2)$ and $N = 6$: $B = +25.0$ ($m = 1$), $+39.2$ ($m = 2$). With the largest admissible m, B/m^2 equals -1.61, -1.26, -0.39, -0.11, $+0.67$, $+0.90$, $+0.92$ for $N = 3, 4, 6, 8, 10, 12, 14$.

The single form ($m = 1$) gives the smallest B; the determinant adds to it. The negative values at small N do not persist. The factors u^{2j} remove the smallness of the forms: at $N = 8$, $\log|\mu_0| = -43.3$ and $\log|\mu_{12}| = +15.8$. Section 11 measures a Hankel construction of a different kind, which proves the irrationality of $\zeta(5)$ [4].

9 Weights and denominators of the moments

The weighted measure of Section 6 is one member of a family, and the family shows where the extra denominators come from.

Lemma 21. *Let w be a real polynomial with $w(0) = 0$ and $w'(\pi/2) = 0$, and put $I_n(w) = \int_0^{\pi/2} w(\theta)\sin^n\theta\,d\theta$. Then $I_1(w) = -I_1(w'')$ and, for odd $n \geq 3$,*

$$n^2 I_n(w) = n(n-1)I_{n-2}(w) - I_n(w'').$$

Proof. Integrate by parts twice with $f = \sin^n$: $\int_0^{\pi/2} wf'' = [wf' - w'f]_0^{\pi/2} + \int_0^{\pi/2} w''f$. The boundary term is $-w'(\pi/2) - w(0)f'(0)$, which vanishes, and $f'' = n(n-1)\sin^{n-2} - n^2\sin^n$ for $n \geq 2$, $f'' = -\sin$ for $n = 1$. □

If w'' satisfies the hypotheses again, and so on down to a constant, then all $I_n(w)$ with n odd are rational, and the θ-degree of w is the weight that the measure adds. In degree two the conditions leave exactly $w_2 = \theta^2 - \pi\theta$. In degree four the conditions $w(0) = 0$, $w'(\pi/2) = 0$, $w''(0) = 0$ and $w'''(\pi/2) = 0$ on a monic quartic leave exactly $w_4 = \theta^4 - 2\pi\theta^3 + \pi^3\theta$. Each application of the recurrence divides the inhomogeneous term by n^2 with n odd.

Observation 22 (Measured; 60 digits)**.** For the measures $|t|\,w(|\arcsin t|)\,dt/\sqrt{1-t^2}$ with $w = 1$, w_2, w_4, the starting value $2\int_0^{\pi/2} w(\theta)\,\mathrm{artanh}(\sin\theta)\,d\theta$ equals $4G$, $-8\beta(4)$ and $96\beta(6)$ respectively, and the powers of 5 and of 7 in the denominators of $I_7(w)$ and $I_9(w)$ are 1, 3 and 5 (the power of 3 in $I_9(w)$ is 2, 6 and 9).

The value for w_2 is Lemma 12. The conditions at $\theta = \pi/2$ remove the boundary terms that otherwise bring lower β-values multiplied by powers of π, as in $\int_0^{\pi/2} \theta^3/\sin\theta\,d\theta = \frac{3}{2}\pi^2 G - 12\beta(4)$; the same integration by parts leaves the inhomogeneous term $I_n(w'')/n^2$, which raises the power of the odd primes in the moments by two for every two units of weight (for 5 and 7 at $n \leq 9$). Two other parametrisations give the same picture:

1. The uniform measure on the segment $i[-1,1]$ has moments $(-1)^k/(2k+1)$, and with the kernel $(-\log s)^2(1-ts)^{-1}$ its starting value is $\beta(4)$ itself. This is the half-integer family of Section 5: the weight comes from the kernel, and an entry of weight four carries p^4 (Observation 9).

2. The moments $c(k)^2$, the moments of a product of two independent arcsine variables, have denominators that are powers of 2; the starting value of weight two is $\sum_k c(k)^2/(2k+1) = 4G/\pi$, and the next two starting values gave no relation with β-, ζ- and logarithmic constants divided by π (PSLQ at 60 and 45 digits).

10 Hypergeometric recurrences

A recurrence with small coefficients whose solutions approximate $\beta(4)$ with positive irrationality exponent, as Apéry's recurrence does for $\zeta(3)$, would prove the irrationality of $\beta(4)$ directly. Since $\beta(4) = {}_5F_4(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},1;\frac{3}{2},\frac{3}{2},\frac{3}{2},\frac{3}{2};-1)$, the contiguity relations of ${}_5F_4$ in its nine parameters give a field of 5×5 matrices (a conservative matrix field in the sense of the Ramanujan Machine project; we use the construction of its `ramanujantools` package, reimplemented without external packages and checked against it), and every lattice direction gives a recurrence. For a direction we walk n steps from the point above, take the quotient p_n/q_n of two entries of the product matrix in lowest terms, and define the exponent δ by $|p_n/q_n - L| = q_n^{-1-\delta}$, L the limit; $\delta > 0$ for all large n proves L irrational.

Observation 23 (Measured)**.** Apéry's recurrence gives $\delta = 0.135$ at $n = 100$. For the 674 directions with steps 0, 1, 2 (the four equal parameters taken as a multiset), and for the 30 best of them from two shifted starting points, no direction has $\delta \geq 0$ at $n = 100$; the median is -0.96 and the best is -0.33. Every limit that could be identified as a quotient of two linear forms (PSLQ, stable at two precisions with at least 120 digits) is an expression in π alone, for example $(\pi - 2)/(8\pi - 12)$ for the best direction; none involves $\beta(4)$ (119 directions with limits in π, 10 unidentified with $\delta \leq -0.70$, 472 with fewer than 120 reliable digits). The analogous scan of ${}_3F_2$ for G gives 13 limits in π and none involving G.

With the projection used here the measurable directions converge to expressions in π, so the scan gives no exponent for $\beta(4)$ itself; a projection that targets $\beta(4)$ was not tried. The very-well-poised forms of Section 4 come from the same hypergeometric family. For 151 parameter vectors η with $3 \leq \eta_0 \leq 8$ and all 11325 pairs, the combination free of $\beta(2)$ has $\log|L_n| + \log \operatorname{den}(L_n)$ increasing with n, by at least 18.6 per unit of n between $n = 8$ and $n = 12$; a proof needs a decrease.

11 Determinants with poles at the negative squares

Fauzan [4] has proved that $\zeta(5)$ is irrational with Hankel determinants of a functional that differs from the measures above in two respects: its moments are rational numbers whose denominators do not grow with the weight, and the constant enters only through the values of the functional at simple poles. Lai [6] used the same method to prove that 1, $\zeta(3)$ and $\zeta(4)$ are linearly independent over $\mathbb{Q}$. Hankel determinants of zeta values are also studied in [5, 7]. The same Hankel-determinant method proves the irrationality of the p-adic zeta values $\zeta_2(7)$ and $\zeta_3(5)$ [8]; there the gain is the p-adic decay of the determinants and their archimedean size enters only as a cost, so the sign of the archimedean size measured in this section does not decide those proofs. We checked the proofs of [4] and [6] and found no error. For [4] the check does not cover its Appendix C, which bounds the irrationality exponent and is not used for the irrationality. Appendix A lists what was checked in each case. In this subsection we measure the construction of [4], a variant of it, and its analogue for $\beta(s)$, with one normalisation for all of them.

Definition 24. Let $s \geq 2$, $K > N \geq 0$, $r \geq 1$ and $1 \leq h \leq K - N$ be integers, let $a_0 < a_1 < \cdots < a_{K-1}$ be positive rational numbers, and put $D_N(t) = \prod_{j<N}(t + a_j^2)$ and $D_K(t) = \prod_{j<K}(t + a_j^2)$. Let μ_X be a $\mathbb{Q}$-linear map from $\mathbb{Q}[t] + \sum_{j<K} \mathbb{Q}\,(t + a_j^2)^{-1}$ to $\mathbb{Q}[X]$. Define

$$\Delta(X) = \det\Big(\mu_X\big(D_N(t)^r\, t^{i+k}/D_K(t)\big)\Big)_{0 \leq i,k < h},$$

(the argument is reduced to a polynomial plus simple fractions by polynomial division) and let $Q = \Delta/c$, where $c > 0$ is the rational number for which Q is a primitive polynomial in $\mathbb{Z}[X]$. We use two functionals.

(Z) For an integer $q \geq 1$, $a_j = 1 + qj$,

$$\mu_X(t^e) = \frac{(2e+s)!\,|B_{2e+2}|}{(s-1)!\,(2e+2)!},$$
$$\mu_X\Big(\frac{1}{t+m^2}\Big) = m^{s-1}(X - H_m^{(s)}) + \frac{1}{2m} - \frac{1}{s-1},$$

with B_k the Bernoulli numbers and $H_m^{(s)} = \sum_{v \leq m} v^{-s}$. The constant is $C = \zeta(s)$. For $s = 5$ and $q = 1$ this is the functional of [4, (2.2), (2.3)].

(B) For s even, $a_j = j + \frac{1}{2}$,

$$\mu_X(t^e) = \frac{(2e+s)!\,(2^{2e+2}-1)\,|B_{2e+2}|}{(s-1)!\,(2e+1)!\,(2e+2)},$$
$$\mu_X\Big(\frac{1}{t+a_j^2}\Big) = 2(2j+1)^{s-1}(-1)^j\Big(X - \sum_{l<j}\frac{(-1)^l}{(2l+1)^s}\Big) - \frac{1}{2j+1}.$$

The constant is $C = \beta(s)$.

The moments in (B) equal $\binom{2e+s}{2e+1}T_{2e+1}/2^{2e+2}$, where $T_{2e+1} = 2^{2e+2}(2^{2e+2}-1)|B_{2e+2}|/(2e+2)$ is a tangent number, an integer; so their denominators are powers of 2. For $s = 2, 4, 6$ and $e < 250$ they divide 2; for $s = 4$ the first moments are $1, \frac{5}{2}, 14, \frac{255}{2}, 1705$. For $s = 5$, the moments in (Z) have no prime $p \geq 7$ in their denominators to a power higher than one [4, Section 4.2].

Lemma 25. *For the functional (B) with s even, let $g(y) = 1/\sinh(\pi y)$ and*

$$w_s(y) = \frac{(-1)^{s-1}y^s g^{(s-1)}(y)}{(s-1)!} \qquad (y > 0).$$

Then $w_s > 0$, and

$$\mu_{\beta(s)}(R) = \int_0^\infty R(y^2)\,w_s(y)\,dy$$

for every R in the domain of μ_X. Consequently the matrix in Definition 24 at $X = \beta(s)$ is positive definite and $\Delta(\beta(s)) > 0$.

Proof. For $y > 0$, $g(y) = 2\sum_{m\geq 0} e^{-(2m+1)\pi y}$, so

$$(-1)^{s-1}g^{(s-1)}(y) = 2\sum_{m\geq 0}\big((2m+1)\pi\big)^{s-1}e^{-(2m+1)\pi y} > 0.$$

Near 0, $g^{(k)}(y) = O(y^{-1-k})$; at infinity every derivative of g decays like $e^{-\pi y}$.

Moments. Integrating by parts $s-1$ times, with boundary terms $O(y^{2e+1})$ at 0 and exponentially small at infinity,

$$\int_0^\infty y^{2e}w_s(y)\,dy = \frac{(2e+s)!}{(s-1)!\,(2e+1)!}\int_0^\infty y^{2e+1}g(y)\,dy.$$

Termwise integration gives $\int_0^\infty y^{2e+1}g(y)\,dy = 2(2e+1)!\,\pi^{-2e-2}(1-2^{-2e-2})\zeta(2e+2)$, and Euler's formula $\zeta(2e+2) = |B_{2e+2}|(2\pi)^{2e+2}/(2(2e+2)!)$ turns this into $(2^{2e+2}-1)|B_{2e+2}|/(2e+2)$.

Poles. Let $a > 0$. For s even, $y^s/(y^2+a^2)$ is a polynomial of degree $s-2$ plus $(-1)^{s/2}a^s/(y^2+a^2)$, and $1/(y^2+a^2) = a^{-1}\operatorname{Re}(a-iy)^{-1}$. Since $(d/dy)^{s-1}(a-iy)^{-1} = (s-1)!\,i^{s-1}(a-iy)^{-s}$ and $i^{s-1} = -i(-1)^{s/2}$,

$$\Big(\frac{d}{dy}\Big)^{s-1}\frac{y^s}{y^2+a^2} = (s-1)!\,a^{s-1}\operatorname{Im}(a-iy)^{-s}.$$

Integrating by parts $s-1$ times (boundary terms $O(y)$ at 0, exponentially small at infinity),

$$\int_0^\infty \frac{w_s(y)}{y^2+a^2}\,dy = a^{s-1}\int_0^\infty \operatorname{Im}(a-iy)^{-s}\,g(y)\,dy.$$

Write $\operatorname{Im}(a-iy)^{-s} = \Gamma(s)^{-1}\int_0^\infty x^{s-1}e^{-ax}\sin(xy)\,dx$. For $x > 0$, $\int_0^\infty \sin(xy)g(y)\,dy = \sum_{m\geq 0} 2x/(x^2+(2m+1)^2\pi^2) = \frac{1}{2}\tanh(x/2)$, by termwise integration and the partial fraction expansion of tanh; and $\tanh(x/2) = 1 + 2\sum_{k\geq 1}(-1)^k e^{-kx}$. The double integral converges absolutely, so

$$\int_0^\infty \operatorname{Im}(a-iy)^{-s}\,g(y)\,dy = \frac{1}{2\Gamma(s)}\int_0^\infty x^{s-1}e^{-ax}\tanh(x/2)\,dx = \sum_{k\geq 0}\frac{(-1)^k}{(k+a)^s} - \frac{1}{2a^s}.$$

At $a = j + \frac{1}{2}$ the sum equals $2^s(-1)^j(\beta(s) - \sum_{l<j}(-1)^l(2l+1)^{-s})$, and $a^{s-1}2^s = 2(2j+1)^{s-1}$, $a^{s-1}/(2a^s) = 1/(2j+1)$. This is the pole value in (B) at $X = \beta(s)$.

Positivity. Both sides are linear in R, so the representation holds on the whole domain. For a real polynomial $p \neq 0$ of degree less than h the quadratic form of the matrix at p is $\int_0^\infty D_N(y^2)^r\, p(y^2)^2\, D_K(y^2)^{-1} w_s(y)\, dy > 0$. □

For (Z) the same argument with $f(y) = 1/(e^{2\pi y} - 1)$, the weight $2(-1)^{s-1}y^s f^{(s-1)}(y)/(s-1)!$ and Hermite's formula for the Hurwitz zeta function [3, 25.11.29] gives the integral representation; for $s = 5$ this is [4, Proposition 2.2].

If $C = a/b$ and $Q(a/b) \neq 0$, then $b^{\deg Q} Q(a/b)$ is a nonzero integer. A construction of this kind proves irrationality when $\log|Q(C)|$ is negative of order h^2 along a sequence of designs and $Q(a/b) \neq 0$ is ensured, as in [4]. We measure

$$\Lambda = \frac{\log|Q(C)|}{h^2}.$$

For the determinants of Section 6 and of [1] we use the same quantity: Q is the primitive integer polynomial proportional to the determinant as a polynomial in the constant, and h is the size of the matrix. For those determinants Λ agrees to the printed digits with the gaps of Observation 17 at the sizes printed there.

Observation 26 (Measured; exact arithmetic)**.** For the functional (Z) with $s = 5$, $q = 1$ and the parameters of [4], $K = 40n$, $N = 3n$, $h = 37n$, $r = 6$:

n	1	2	3
$\log Q(\zeta(5))$	−265.1	−833.3	−1804.0
$\log Q(\zeta(5))/K^2$	−0.166	−0.130	−0.125

The values are positive, as [4, Proposition 2.2] implies, and these polynomials exclude $\zeta(5) = a/b$ for $b < 1.3 \cdot 10^3$, $7.8 \cdot 10^4$, $1.1 \cdot 10^7$. The bound proved in [4, (7.1), (7.2)], with the cutoff $M = 200$ used there, is $\log Q(\zeta(5))/K^2 \leq -0.0174$ asymptotically.

Observation 27 (Measured; exact arithmetic)**.** With $N = 0$, $r = 1$ and $h = K$, the values of Λ are:

	$\zeta(3)$	$\zeta(5)$	$\zeta(7)$	$\beta(2)$	$\beta(4)$	$\beta(6)$
$K = 12$	−1.05	−0.25	+0.52	+0.10	+1.53	+2.97
$K = 24$	−0.96	−0.09	+0.85	+0.27	+2.03	+3.66

With the parameters of [4] at $n = 1$ ($K = 40$, $N = 3$, $h = 37$, $r = 5$) the functional (B) for $\beta(4)$ gives $\Lambda = +2.88$. Over 33 further choices of (N, r, h) at $K = 24$ with $0 \leq N \leq 9$, $1 \leq r \leq 8$, $6 \leq h \leq 18$, $\log Q(\beta(4))/K^2$ is at least $+1.1$.

The values grow with the weight in both families, by 0.8 to 0.9 for each step of two in s for ζ and by 1.4 to 1.8 for β, and the family of β starts higher: $\beta(2)$ is positive where $\zeta(3)$ is at -1. The next observation isolates one cause.

Observation 28 (Measured; exact arithmetic)**.** For the functional (Z) with only every q-th pole, $N = 0$, $r = 1$, $h = K = 24$:

	$q = 1$	$q = 2$	$q = 3$
$\zeta(3)$	−0.96	−0.14	+0.25
$\zeta(5)$	−0.09	+1.68	+2.99

With the parameters of [4] at $n = 1$, $\log Q(\zeta(5))/K^2$ is -0.166 for $q = 1$ and $+1.35$ for $q = 2$.

In (Z) the poles have spacing 1 in the variable $y = \sqrt{t}$ and the weight decays like $e^{-2\pi y}$; in (B) the spacing is again 1 and the weight decays like $e^{-\pi y}$. Scaling y by 2 shows that (B) has the geometry of (Z) with $q = 2$, and at $K = 24$ the values of $\beta(2)$ and $\beta(4)$ ($+0.27$, $+2.03$) are within 0.41 of those of $\zeta(3)$ and $\zeta(5)$ with $q = 2$ (-0.14, $+1.68$), compared with 1.23 and 2.12 for $q = 1$. The period of the weight in (B) comes from the character modulo 4, and we do not know a representation of $\beta(s)$ of this kind with denser poles. Hermite's formula at $a = j + \frac{1}{4}$ and $a = j + \frac{3}{4}$ has poles at spacing $\frac{1}{2}$ and the weight $e^{-2\pi y}$, which is the same ratio after scaling, and its values contain $\zeta(s, \frac{1}{4}) + \zeta(s, \frac{3}{4}) = (4^s - 2^s)\zeta(s)$ besides $\beta(s)$.

Observation 29 (Measured; exact arithmetic)**.** On the common scale Λ:

weight	construction	Λ (size h in brackets)
two	[1], ratios $48 : 7 : 4 : 4 : 2$	-0.30 (12), -0.22 (20), -0.15 (28), -0.12 (40)
two	arcsine measure, rows and columns of Δ_n^ν (Observation 17)	-0.16 (6), $+0.04$ (10), $+0.10$ (12)
two	(B), $s = 2$	$+0.10$ (12), $+0.27$ (24)
four	Δ_n^ν (Observation 17)	$+2.09$ (6), $+2.75$ (10), $+2.94$ (12)
four	(B), $s = 4$	$+1.53$ (12), $+2.03$ (24), $+2.88$ (37)

For (B) with $s = 2$, $K = 24$, over 27 choices with $0 \le N \le 6$, $1 \le r \le 5$, $12 \le h \le 24$, every value is positive and the smallest is $+0.27$.

At weight two both families with plain rows lie between -0.2 and $+0.3$, and the rows of [1] lower the value by about 0.4 (Observation 18 locates this at primes above n). At weight four both families start at $+2$ or above.

12 The trade-off

Across the tests one pattern holds. The arcsine measure keeps denominators small and ties $\beta(4)$ to $\operatorname{Im} \operatorname{Li}_4((1 + i)/2)$ (Sections 2–3). The weighted arcsine measure of Section 6 separates $\beta(4)$, with entries in $\mathbb{Q} + \mathbb{Q}\beta(4)$ (Proposition 16), but its moments carry the squares of the odd numbers (Lemma 11) and its determinants exceed the Catalan determinants with the same rows by about 3.3 per n^2 at $n = 32$ (Observation 17). The half-integer determinants exceed theirs by more. In each case the size of the determinants is of the same order at both weights, while the denominators follow the weight: at every odd prime the power per entry doubles (Observation 18), and Section 9 locates the extra denominators in the moment recurrence (Lemma 21). The recurrence scan of Section 10 found no direction with positive exponent. A determinant proof for $\beta(4)$ along the lines of [1] would need a construction in which the size decreases with the weight as fast as the denominators grow, as in the integrals of Apéry and Beukers for $\zeta(3)$, where each further integration variable contributes to both. The determinants of [4] are a construction of this kind at weight five: the denominators of weight five stay in the values at the poles, and the measured value stays negative (Observation 26). Their analogue for $\beta(4)$ stays near $+2$ (Observation 27); the character modulo 4 halves the density of the poles relative to the decay of the weight (Observation 28). We do not know a construction for $\beta(4)$ that has the denominators of [4] and the density of poles of ζ.

Code and data

The scripts, their logs and the source of this note are in the folder `beta4/` of the repository `https://github.com/MarcosCarreira/catalan-designs`. The subfolder `beta4/scouts/` holds the exploration scripts that preceded the measurements; they have no pass criteria and no logs.

Acknowledgement

This note was prepared with the assistance of Claude (Anthropic). All computations were run by the author on his own computer; the logs are listed in Appendix B.

References

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[8] I. Kilani, *On the irrationality of* $\zeta_3(5)$, preprint, October 2026, doi:10.5281/zenodo.23203596; Lean sources at `https://github.com/Ijtihed/zeta3-5-lean`. The proof for $\zeta_2(7)$ is cited there.

[9] M. Costa Santos Carreira, *General designs and an irrationality exponent for Catalan's constant*, preprint, Zenodo, 12 October 2026, doi:10.5281/zenodo.23326324; scripts and logs at `https://github.com/MarcosCarreira/catalan-designs`.

Appendices

Appendix A. Checks of two proofs

This appendix lists the checks summarised in Section 11. The scripts and runs are listed in Appendix B.

A.1. The proof of Fauzan

For the proof that $\zeta(5)$ is irrational [4]: Sections 2 to 7 were read in full, Proposition 4.1 line by line. The distribution formula of [4, Lemma 3.2] was checked p-adically for $p = 7, 11, 13$. The constants of [4, Appendices A and B] were recomputed: the two integrals of Appendix B and the constant A^* agree exactly in rational arithmetic, and the two inequalities of [4, Lemma 6.1] hold with margins 0.005 and $6 \cdot 10^{-8}$. On the exact determinants for $n = 1, 2, 3$ the bounds [4, (3.12)] and [4, Proposition 4.3] hold at every prime, and the bound of Proposition 4.3 is attained at some prime for each n. We found no error. Appendix C of [4], which bounds the irrationality exponent and is not used for the irrationality, was not checked.

A.2. The proof of Lai

For the proof that 1, $\zeta(3)$ and $\zeta(4)$ are linearly independent over $\mathbb{Q}$ [6]: we checked Lemma 2.2, Lemmas 4.1 and 4.4 on the exact determinants for $8 \le n \le 24$, the integrals and finite checks of Section 5, and the constants, Lemmas 6.1 and 6.2 and the identity (6.9) of Section 6, and found no error.

Appendix B. Scripts and runs

All scripts are written in Python and take no arguments; their settings are constants at the top of each file. Each script states its pass criteria before running and ends by printing the number of failed criteria, and each of the runs listed here ended with zero failures. All runs were made by the author on his own computer on 7 and 8 October 2026.

script	content	log
`b4_balance.py`	half-integer determinants, both weights (Observation 8)	`_20261007_202313`
`b4_denominators.py`	denominators with and without the cancellation (Observation 9)	`_20261007_203834`
`b4_route3.py`	weighted arcsine measure: gaps of Δ_n^ν, of the Catalan and [1] determinants, denominators by prime, Hankel determinants	`_20261008_110012`
`b4_cmf_scan.py`	hypergeometric recurrences and very-well-poised pairs (Section 10)	`_20261008_111915`
`b4_cmf_filter.py`	limits of the recurrences (Observation 23)	`_20261008_115501`
`fz5_check.py`	positive control: the determinants of [4] for $n = 1, 2, 3$ (Observation 26)	`_20261008_150759`
`b4_fauzan.py`	Observations 27–29	`_20261008_153547`
`fz5_padic_check.py`	checks of [4, (3.12), Proposition 4.3] on the exact determinants, $n = 1, 2, 3$ (Section 11)	`_20261008_160037`
`audit_fauzan_lai.py`	checks of the constants and lemmas of [4] and [6] listed in Appendix A (scripts in `fauzan_audit/`, `lai_audit/`)	`_20261008_163817`